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1. In the expression $Y + X'$, Y , which operator will be evaluated first?

(a) ' (b) + (c) . (d) ,

Computer – Boolean Algebra (Operator Precedence) – 2021

2. Which of the following is false?

(a) $x + y = y + x$ (b) $x \cdot y = y \cdot x$
(c) $x \cdot x' = 1$ (d) $x + x' = 1$

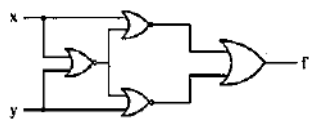
Computer – Boolean Algebra Laws – 2021

3. A Boolean function $x'y' + xy + x'y$ is equivalent to

(a) $x' + y'$ (b) $x + y$
(c) $x + y'$ (d) $x' + y$

Computer – Boolean Expression Simplification – 2021

4. Which of the following logic operations is performed by the following given combinational circuit?



(a) Exclusive-OR (b) Exclusive-NOR
(c) NAND (d) NOR

Computer – Logic Gates – 2021

5. The output of an AND gate with three inputs A, B and C is HIGH when

(a) $A = 1, B = 1, C = 0$ (b) $A = 0, B = 0, C = 0$
(c) $A = 1, B = 1, C = 1$ (d) $A = 1, B = 0, C = 1$

6. Don't care conditions can be used for simplifying Boolean expressions in

(a) registers (b) terms
(c) K-maps (d) latches

Computer – Karnaugh Map – 2021

7. In the expression $A + BC$, the total numbers of minterms will be

(a) 2 (b) 3
(c) 4 (d) 5

Computer – Boolean Minterms – 2021

8. Given the following truth table:

A	B	x
0	0	1
0	1	0
1	0	0
1	1	0

Which of the following Boolean functions does it represent?

(a) OR (b) XOR
(c) NOR (d) XNOR

COMPUTER – Boolean Logic– 2022

9. The simplified form of Boolean expression $AB + AB'$ is

(a) A (b) B (c) $1 + A$ (d) $1 + B'$

COMPUTER – Boolean Algebra– 2022

10. Match List I with List II

	List I		List II
A.	$(x + y)'$	I.	I
B.	$x + 1$	II.	$x' + y'$
C.	$(xy)'$	III.	$x' \cdot y'$
D.	$x + 0$	IV.	x

Choose the correct answer from the options given below:

(a) A – II; B – I; C – III; D – IV
(b) A – II; B – IV; C – I; D – III
(c) A – III; B – I; C – IV; D – II
(d) A – III; B – I; C – II; D – IV

COMPUTER – Boolean Identities– 2022

11. The simplified form of Boolean Expression:

$\bar{A}BC + ABC$ is _____.

(a) AC (b) AB
(c) BC (d) 0

COMPUTER – Boolean Algebra– 2022

12. If $\oplus$ and $\odot$ denote the exclusive OR and exclusive NOR operations, respectively, then which one of the following is not correct?

(a) $\bar{P} \oplus \bar{Q} = P \odot Q$
(b) $\bar{P} \oplus Q = P \odot Q$
(c) $\bar{P} \oplus \bar{Q} = \bar{P} \odot Q$
(d) $(P \oplus \bar{P}) + Q = (P \odot \bar{P}) \odot \bar{Q}$

COMPUTER – Boolean Algebra – 2023

13. If x, w, y, z are Boolean variables, then which of the following is incorrect?

(a) $wx + w(x + y) + x(x + y) = x + wy$
(b) $w\bar{x}(y + \bar{z}) + \bar{w}x = \bar{w} + x + \bar{y}z$
(c) $(w\bar{x}(y + x\bar{z}) + \bar{w}\bar{x})y = x\bar{y}$
(d) $(w + y)(wxy + wyz) = wxy + wyz$

COMPUTER – Boolean Algebra – 2023

14. The number of possible Boolean function that can be defined for n Boolean variables over n-valued Boolean algebra is _____.

(a). n^{2^n} (b). 2^{n^2}
(c). 2^{2^n} (d). n^{n^n}

COMPUTER – Boolean Function – 2023

15. The function represented by the k-map given below is

k-map				
C/AB	00	01	11	10
0	0	1	1	1
1	0	0	1	0

(a) $BC + AB + AC'$ (b) $BC' + AB + AC'$
(c) $(B \oplus C)'$ (d) $A \cdot BC$

(CUET-PG 2024) Digital Logic – K-Map Simplification





16. Let A' represent complement of A. Which of the following Boolean expression is/are true?

- (A) $A + AB = A$ (B) $(A + B)' = A'B'$
(C) $(A')' = A$ (D) $(AB)' = A' + B'$

Choose the correct answer from the options given below:

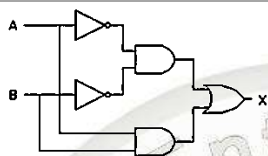
- (a) A, B and D only (b) A and D only
(c) A, B, C and D (d) B and D only

(CUET-PG 2024) Boolean Algebra – Laws & Identities

17. The following circuit generates the same output as?

(CUET-PG 2025) Logic Gate

- (a) XOR GATE
(b) XNOR GATE
(c) NAND GATE
(d) NOR GATE



18. Consider the following Karnaugh Map (k-map). Minimal Function generated by this Karnaugh map is

RS \ PQ	PQ			
	00	01	11	10
00	1	1		1
01	X			
11	X			
10	1	1		X

(CUET-PG 2025) K- map

- (a) $Q'S + P'Q$ (b) $Q'S' + P'S'$
(c) $P'Q' + P'S' + Q'S'$ (d) $PQ + Q'S' + P'S'$

19. What should be the output of the following boolean expression after simplifying it to a minimum number of variables?

$$a'b' + ab + a'b$$

(CUET-PG 2025) Boolean Algebra

- (a) $a + b'$ (b) $a' + b$
(c) $a' + b'$ (d) $a + b$

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	c	d	a	c	c	d	c	a	d
11.	12.	13.	14.	15.	16.	17.	18.	19.	
c	d	c	c	b	c	b	b	b	

